
SUMMARY

The central purpose of The Financial Manager's Decisions is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

General Motors in the 1920s: Donaldson Brown, who had come from DuPont, built financial controls linking return on investment to decisions. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

02.T1 name and tell apart the three decisions using a new business

02.T2 state the work of a treasurer and a controller

02.T3 build a balance sheet from a new list of items and check that it balances

PAUSE AND CHECK 02.P1 After the three decisions, state the rule in your own words.

PAUSE AND CHECK 02.P2 After the balance sheet, state the rule in your own words.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

AN ELEMENTARY INTRODUCTION TO MICROECONOMICS



KUTUSH KUMAR, KUTUM PAKHI

later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to the financial manager's decisions. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For the three big decisions, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

2.2 WHAT A FINANCIAL MANAGER DOES

The purpose here is to make what a financial manager does concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Dividend decision** means A choice about how much profit to pay out to owners and how much to keep for the business. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Treasurer** means The officer in charge of handling cash, raising money and keeping relations with banks. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Controller** means The officer in charge of accounting records, reports and checking that money is used as planned. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Financial planning** means Estimating the money a business will need and receive in future, and arranging how to meet the gap. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of what a financial manager does before moving to application.

Dividend decision. A choice about how much profit to pay out to owners and how much to keep for the business..

The term dividend decision gives the chapter a precise name for this part of the problem.

Treasurer. The officer in charge of handling cash, raising money and keeping relations with banks..

The term treasurer gives the chapter a precise name for this part of the problem.

Controller. The officer in charge of accounting records, reports and checking that money is used as planned..

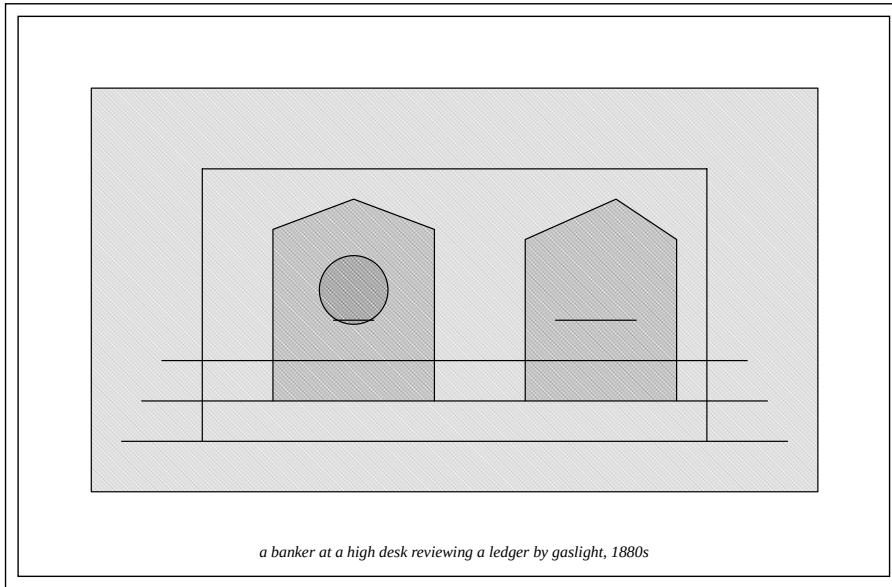
The term controller gives the chapter a precise name for this part of the problem.

Financial planning. Estimating the money a business will need and receive in future, and arranging how to meet the gap..

The term financial planning gives the chapter a precise name for this part of the problem.

The central question in what a financial manager does is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.



CHAPTER II

The Financial Manager's Decisions

A practical question about the financial manager's decisions



This chapter introduces the financial manager's decisions for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

XII. VALUATION OF BONDS	91
12.1 The idea of valuation	92
12.2 What a bond is	93
12.3 Pricing a bond	94
12.4 Yield and price	95
XIII. VALUATION OF SHARES	99
13.1 Common stock	100
13.2 Valuing preferred stock	101
13.3 Valuing common stock	101
XIV. WORKING CAPITAL AND SHORT-TERM FINANCING	107
14.1 Working capital and how to finance it	108
14.2 Sources of short-term finance	109
14.3 Cycles, float and estimating needs	110
XV. LONG-TERM FINANCING AND THE CAPITAL MARKET	117
15.1 Term loans and the features of financing	118
15.2 Debt and equity as sources	119
15.3 Raising funds in the capital market	120
15.4 The cost of capital	120
XVI. LEASING	125
16.1 What leasing is	126
16.2 Kinds of lease	127
16.3 The lease-or-buy decision	127
APPENDIX A. MATHS YOU WILL USE	133
A.1 A1.1 Percentages and percentage change	134
A.2 A1.2 Ratios and averages	134
A.3 A1.3 Rearranging an equation and rounding	134
A.4 A1.4 Reading a graph	134
GLOSSARY	137
ANSWERS TO QUESTIONS	145
INDEX	153

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

SUMMARY

The central purpose of What Finance Is is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

The chapter has connected its main terms, method, and business decision.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

Finance becomes stronger when a number is followed by a question about meaning.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

Most errors arise from mixing terms, timing, signs, or decision rules.

IN HISTORY

The Dutch East India Company (VOC), chartered 1602, sold shares to the public that traded in Amsterdam, an early company separating owners from managers. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

A historical case shows how the chapter's financial problem appeared in practice.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

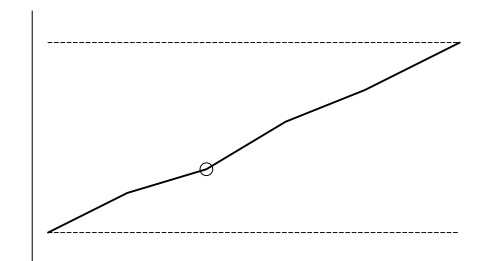


FIG. 1.1. What Finance Is figure 1

1.1 FINANCE IN PLAIN WORDS

The purpose here is to make finance in plain words concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Finance** means The activity of finding money, using it, and judging whether the result was worth it. A shop owner who borrows to buy stock does finance. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Cash flow** means Money that actually moves into or out of a business. Paying a supplier Tk 5,000 is a cash flow of minus Tk 5,000. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Profit** means What a business earns from sales after subtracting the costs of earning them. Sales of Tk 100 and costs of Tk 80 leave profit of Tk 20. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

This section builds the idea of finance in plain words before moving to application.

Finance. The activity of finding money, using it, and judging whether the result was worth it.

The term finance gives the chapter a precise name for this part of the problem.

Cash flow. Money that actually moves into or out of a business.

The term cash flow gives the chapter a precise name for this part of the problem.

Profit. What a business earns from sales after subtracting the costs of earning them.

The term profit gives the chapter a precise name for this part of the problem.

The central question in finance in plain words is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

SUMMARY

The central purpose of Present Value and Discounting is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

The chapter has connected its main terms, method, and business decision.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

Finance becomes stronger when a number is followed by a question about meaning.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

Most errors arise from mixing terms, timing, signs, or decision rules.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

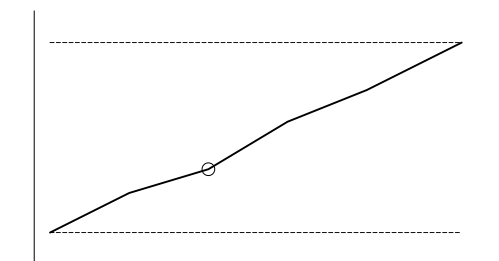


FIG. 3.1. *The Goal of the Firm and Agency Theory figure 1*

3.1 PROFIT MAXIMISATION AND ITS LIMITS

The purpose here is to make profit maximisation and its limits concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Profit maximisation** means Aiming to make the largest profit possible, usually within one year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Wealth maximisation** means Aiming to raise the value of the owners' investment over time, taking account of when money arrives and how risky it is. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Shareholder** means A person who owns part of a company by holding its shares. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Share price** means The price at which one share of a company can be bought or sold today. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

This section builds the idea of profit maximisation and its limits before moving to application.

Profit maximisation. Aiming to make the largest profit possible, usually within one year..

The term profit maximisation gives the chapter a precise name for this part of the problem.

Wealth maximisation. Aiming to raise the value of the owners' investment over time, taking account of when money arrives and how risky it is..

The term wealth maximisation gives the chapter a precise name for this part of the problem.

Shareholder. A person who owns part of a company by holding its shares..

The term shareholder gives the chapter a precise name for this part of the problem.

Share price. The price at which one share of a company can be bought or sold today..

The term share price gives the chapter a precise name for this part of the problem.

The central question in profit maximisation and its limits is what changes when the underlying decision changes.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For discounting a single sum, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

5.2 MIXED STREAMS OF CASH FLOWS

The purpose here is to make mixed streams of cash flows concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Discount rate** means The yearly percentage used in discounting; it states the return you could earn elsewhere. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Cash inflow** means Money coming into a business or to an investor, shown as a positive number. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For mixed streams of cash flows, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

The central question in discounting a single sum is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of mixed streams of cash flows before moving to application.

Discount rate. The yearly percentage used in discounting; it states the return you could earn elsewhere..

The term discount rate gives the chapter a precise name for this part of the problem.

Cash inflow. Money coming into a business or to an investor, shown as a positive number..

The term cash inflow gives the chapter a precise name for this part of the problem.

The central question in mixed streams of cash flows is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

The business meaning matters as much as the arithmetic.

This section builds the idea of owners, managers and agency theory before moving to application.

Agency cost. A cost the principal bears to reduce the agency problem or from the loss it still causes..

The term agency cost gives the chapter a precise name for this part of the problem.

Incentive. A reward, such as a bonus, that encourages a person to act in a chosen way..

The term incentive gives the chapter a precise name for this part of the problem.

The business meaning matters as much as the arithmetic.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For wealth maximisation, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

3.3 OWNERS, MANAGERS AND AGENCY THEORY

The purpose here is to make owners, managers and agency theory concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Agency cost** means A cost the principal bears to reduce the agency problem or from the loss it still causes. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Incentive** means A reward, such as a bonus, that encourages a person to act in a chosen way. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

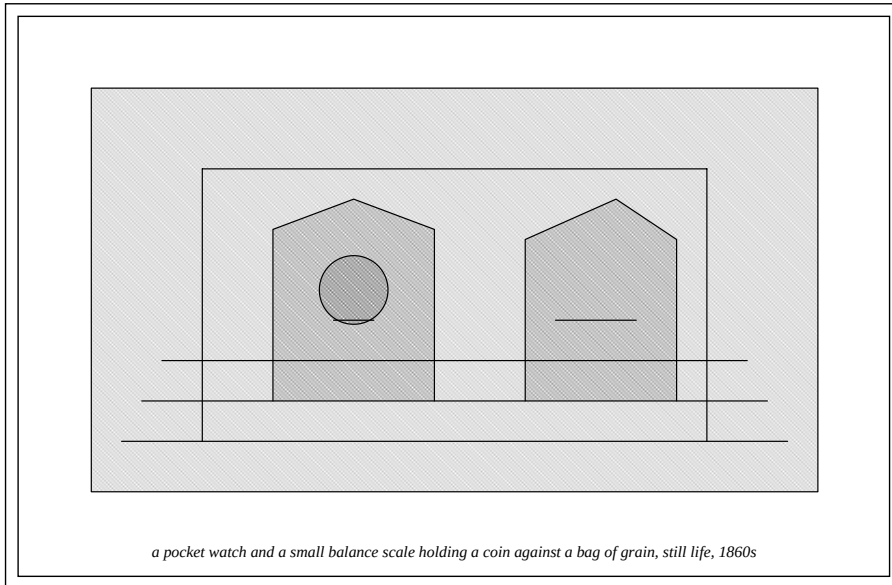
Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For owners, managers and agency theory, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to the goal of the firm and agency theory. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the



CHAPTER V

Present Value and Discounting

A practical question about present value and discounting



This chapter introduces present value and discounting for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and

the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

SUMMARY

The central purpose of Interest and Future Value is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

SOURCES

Chapter 1, 'What Finance Is', section 1.1 'Finance in plain words' — profit as defined there; Chapter 5, 'Present Value and Discounting', section 5.1 'Discounting a single sum' — how timing is measured in wealth..

REVIEW QUESTIONS

03.T1 explain why the aim of the largest profit can mislead, using new yearly figures

03.T2 define wealth maximisation and say how it differs from profit maximisation

03.T3 describe an agency problem and two remedies in a hospital-equipment firm

PAUSE AND CHECK 03.P1 After profit maximisation, state the rule in your own words.

PAUSE AND CHECK 03.P2 After the agency relationship, state the rule in your own words.

In this chapter, **Factor table** means A table of ready-made multipliers that saves repeating a long calculation. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For future value of a single sum, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to interest and future value. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ $(1 + \text{rate})$. Steps: Tk 100,000 $\times$ $(1 + 0.10)$ = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when

Factor table. A table of ready-made multipliers that saves repeating a long calculation..

The term factor table gives the chapter a precise name for this part of the problem.

The central question in future value of a single sum is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

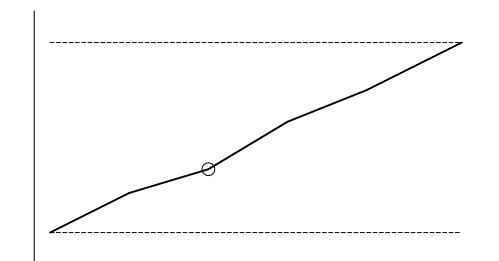


FIG. 4.1. Interest and Future Value figure 1

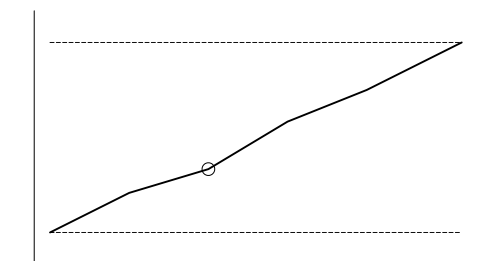


FIG. 4.2. Interest and Future Value figure 2

4.1 WHY TIME CHANGES THE VALUE OF MONEY

The purpose here is to make why time changes the value of money concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Time value of money** means The idea that a taka received today is worth more than a taka received later. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Opportunity cost** means What you give up by choosing one use of money instead of the best alternative. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of why time changes the value of money before moving to application.

Time value of money. The idea that a taka received today is worth more than a taka received later..

The term time value of money gives the chapter a precise name for this part of the problem.

Opportunity cost. What you give up by choosing one use of money instead of the best alternative..

The term opportunity cost gives the chapter a precise name for this part of the problem.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

SUMMARY

The central purpose of Evaluating Projects: Payback and Net Present Value is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

IN HISTORY

A historical case shows how the chapter's financial problem appeared in practice.

Irving Fisher, The Theory of Interest (1930), which states the discounting idea in its modern form. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

05.T1 explain discounting as the reverse of compounding using new figures

05.T2 find the present value of a single sum for two discount rates

05.T3 find present value of a mixed stream and compare two streams

PAUSE AND CHECK 05.P1 After present value of a single sum, state the rule in your own words.

PAUSE AND CHECK 05.P2 After the mixed stream, state the rule in your own words.

to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to evaluating projects: payback and net present value. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

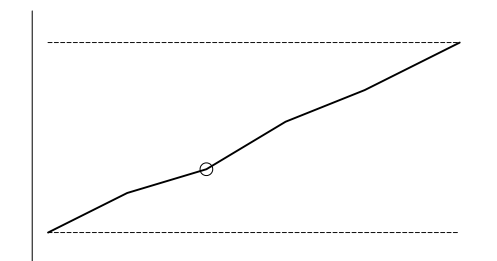


FIG. 6.1. Annuities figure 1

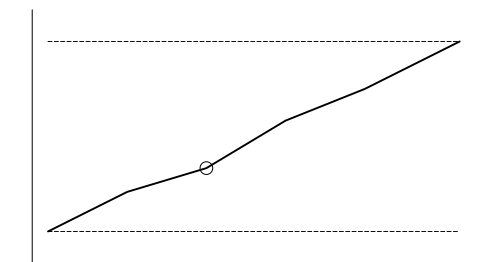


FIG. 6.2. Annuities figure 2

6.1 WHAT AN ANNUITY IS

The purpose here is to make what an annuity is concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Annuity** means A series of equal payments made at equal intervals for a fixed number of periods. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

This section builds the idea of what an annuity is before moving to application.

Annuity. A series of equal payments made at equal intervals for a fixed number of periods..

The term annuity gives the chapter a precise name for this part of the problem.

The central question in what an annuity is what changes when the underlying decision changes.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For payback period, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

8.2 NET PRESENT VALUE

The purpose here is to make net present value concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Net present value** means The present value of all a project's future cash flows minus its initial investment. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Internal rate of return** means The discount rate at which a project's net present value is exactly zero. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For net present value, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

The central question in payback period is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of net present value before moving to application.

Net present value. The present value of all a project's future cash flows minus its initial investment..

The term net present value gives the chapter a precise name for this part of the problem.

Internal rate of return. The discount rate at which a project's net present value is exactly zero..

The term internal rate of return gives the chapter a precise name for this part of the problem.

The central question in net present value is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

Annuity due. An annuity whose payments fall at the start of each period..

The term annuity due gives the chapter a precise name for this part of the problem.

The central question in present value of an annuity is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of solving for the payment before moving to application.

Instalment. One of a series of regular payments used to repay a loan or build a sum..

The term instalment gives the chapter a precise name for this part of the problem.

The central question in solving for the payment is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

In this chapter, **Annuity due** means An annuity whose payments fall at the start of each period. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For present value of an annuity, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

6.4 SOLVING FOR THE PAYMENT

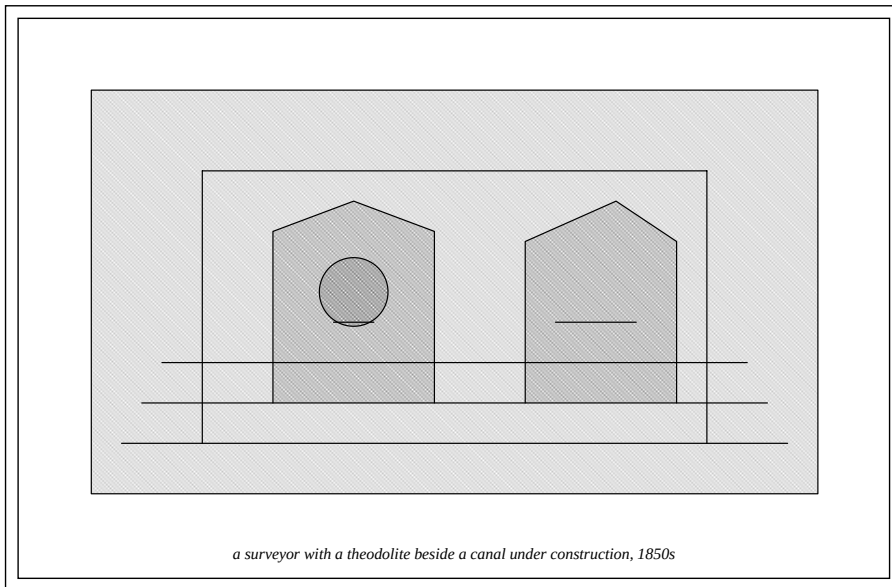
The purpose here is to make solving for the payment concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Instalment** means One of a series of regular payments used to repay a loan or build a sum. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For solving for the payment, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.



CHAPTER VIII

Evaluating Projects: Payback and Net Present Value

A practical question about evaluating projects: payback and net present value



This chapter introduces evaluating projects: payback and net present value for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because

the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

SUMMARY

The central purpose of Capital Budgeting: Process and Cash Flows is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

The Concorde programme: UK and France agreed in 1962 to build it and continued after costs soared, a famous example of sunk-cost thinking; service ended in 2003. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

- 07.T1 say why capital budgeting matters to a business, with a new example
- 07.T2 list the steps of the process in order and what each step produces
- 07.T3 sort a list of costs into relevant and sunk

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

each method begins with a disciplined description of the cash or value being studied.

SUMMARY

The central purpose of Annuities is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Johan de Witt, 1671 report to the States of Holland on valuing life annuities for the Dutch Republic. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

- 06.T1 say what makes a series of payments an annuity and name its two timing forms
- 06.T2 find the future value of an annuity for several rate and period pairs as ordinary and as due
- 06.T3 find the present value of a new instalment plan
- 06.T4 compute the yearly deposit needed for a stated goal
- PAUSE AND CHECK 06.P1 After annuity types, state the rule in your own words.
- PAUSE AND CHECK 06.P2 After future value of an ordinary annuity, state the rule in your own words.

the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

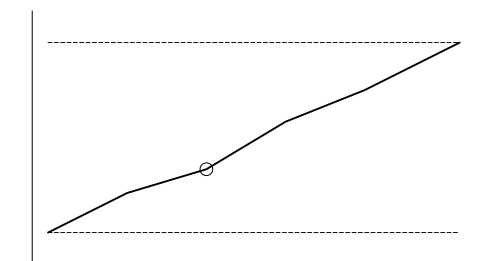


FIG. 7.1. Capital Budgeting: Process and Cash Flows figure 1

7.1 WHY FIRMS BUDGET FOR CAPITAL

The purpose here is to make why firms budget for capital concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Capital budgeting** means Planning and choosing long-lasting investments by comparing the money they cost with the money they should bring. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Capital expenditure** means Spending on an asset that will help a business for more than one year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Independent project** means A project that can be accepted or rejected without affecting any other project. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Mutually exclusive projects** means Projects of which only one can be chosen, because choosing one rules out the other. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared.

This section builds the idea of why firms budget for capital before moving to application.

Capital budgeting. Planning and choosing long-lasting investments by comparing the money they cost with the money they should bring..

The term capital budgeting gives the chapter a precise name for this part of the problem.

Capital expenditure. Spending on an asset that will help a business for more than one year..

The term capital expenditure gives the chapter a precise name for this part of the problem.

Independent project. A project that can be accepted or rejected without affecting any other project..

The term independent project gives the chapter a precise name for this part of the problem.

Mutually exclusive projects. Projects of which only one can be chosen, because choosing one rules out the other..

The term mutually exclusive projects gives the chapter a precise name for this part of the problem.

The central question in why firms budget for capital is what changes when the underlying decision changes.

later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

11.4 EFFICIENT MARKETS

The purpose here is to make efficient markets concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Capital asset pricing model** means A formula for required return: risk-free rate plus beta times the market risk premium. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Security market line** means The graph of the capital asset pricing model, with beta on one axis and required return on the other. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Efficient market** means A market in which prices already reflect the available information. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For efficient markets, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to required return, capm and market efficiency. Start by listing the known amounts and their dates. Separate cash inflows from

This section builds the idea of efficient markets before moving to application.

Capital asset pricing model. A formula for required return: risk-free rate plus beta times the market risk premium..

The term capital asset pricing model gives the chapter a precise name for this part of the problem.

Security market line. The graph of the capital asset pricing model, with beta on one axis and required return on the other..

The term security market line gives the chapter a precise name for this part of the problem.

Efficient market. A market in which prices already reflect the available information..

The term efficient market gives the chapter a precise name for this part of the problem.

The central question in efficient markets is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

WHERE PEOPLE GO WRONG

Most errors arise from mixing terms, timing, signs, or decision rules.

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

A historical case shows how the chapter's financial problem appeared in practice.

The Channel Tunnel, opened 6 May 1994, whose traffic forecasts and costs differed greatly from plan. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

08.T1 compute payback and net present value for new project data and state the decision rule for each

08.T2 say what the internal rate of return means in plain words

08.T3 rank three new projects by profitability index

PAUSE AND CHECK 08.P1 After payback, state the rule in your own words.

PAUSE AND CHECK 08.P2 After net present value, state the rule in your own words.

In this chapter, **Risk premium** means The extra return demanded for taking risk instead of a risk-free investment. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For risk premium and required return, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

11.2 DIVERSIFICATION AND BETA

The purpose here is to make diversification and beta concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Portfolio** means A collection of investments held together by one investor. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Diversification** means Spreading money across many investments so that one failure does less damage. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Systematic risk** means Risk that affects all investments, such as a recession; it cannot be diversified away. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Risk premium. The extra return demanded for taking risk instead of a risk-free investment..

The term risk premium gives the chapter a precise name for this part of the problem.

The central question in risk premium and required return is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of diversification and beta before moving to application.

Portfolio. A collection of investments held together by one investor..

The term portfolio gives the chapter a precise name for this part of the problem.

Diversification. Spreading money across many investments so that one failure does less damage..

The term diversification gives the chapter a precise name for this part of the problem.

Systematic risk. Risk that affects all investments, such as a recession; it cannot be diversified away..

The term systematic risk gives the chapter a precise name for this part of the problem.

The central question in diversification and beta is what changes when the underlying decision changes.

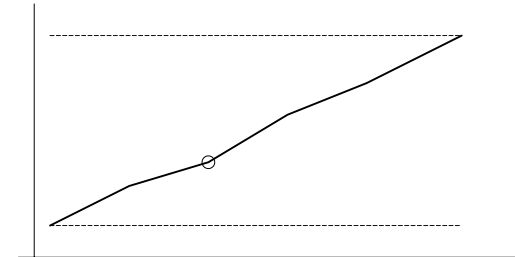


FIG. 9.1. Risk and Return figure 1

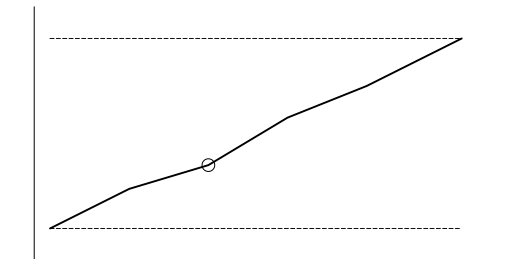


FIG. 9.2. Risk and Return figure 2

9.1 RETURN AND RISK

The purpose here is to make return and risk concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Return** means The gain or loss from an investment, including income received and any change in value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Rate of return** means The return in one period divided by the sum invested, written as a percentage. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

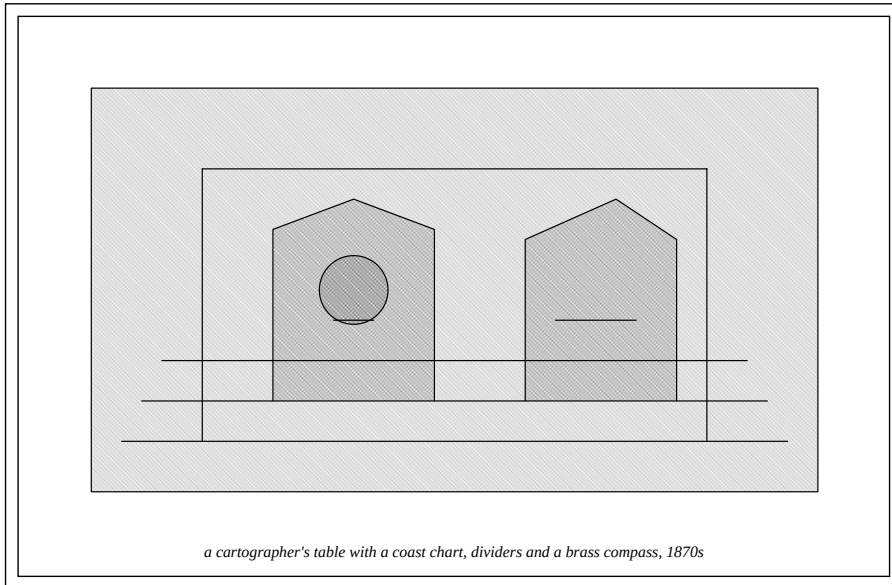
This section builds the idea of return and risk before moving to application.

Return. The gain or loss from an investment, including income received and any change in value..

The term return gives the chapter a precise name for this part of the problem.

Rate of return. The return in one period divided by the sum invested, written as a percentage..

The term rate of return gives the chapter a precise name for this part of the problem.



CHAPTER XI

Required Return, CAPM and Market Efficiency

A practical question about required return, capm and market efficiency



This chapter introduces required return, capm and market efficiency for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

Expected value. The average of possible results, each weighted by its probability..

The term expected value gives the chapter a precise name for this part of the problem.

Variance. The probability-weighted average of the squared gaps between each result and the expected value..

The term variance gives the chapter a precise name for this part of the problem.

The central question in describing uncertainty is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

In this chapter, **Expected value** means The average of possible results, each weighted by its probability. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Variance** means The probability-weighted average of the squared gaps between each result and the expected value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For describing uncertainty, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

9.3 MEASURING RISK

This section builds the idea of measuring risk before moving to application.

Standard deviation. The square root of the variance; it shows how far results typically stray from the expected value..

The term standard deviation gives the chapter a precise name for this part of the problem.

Coefficient of variation. Standard deviation divided by expected value, giving risk per unit of expected return..

The term coefficient of variation gives the chapter a precise name for this part of the problem.

The central question in measuring risk is what changes when the underlying decision changes.

The purpose here is to make measuring risk concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Standard deviation** means The square root of the variance; it shows how far results typically stray from the expected value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Coefficient of variation** means Standard deviation divided by expected value, giving risk per unit of expected return. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared.

the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

SUMMARY

The central purpose of Leverage is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

10.4 TOTAL LEVERAGE AND RISK

The purpose here is to make total leverage and risk concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Earnings per share** means Earnings left for owners of common stock divided by the number of shares. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Preferred stock** means Shares that pay a fixed dividend, which must be paid before any dividend on common stock. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For total leverage and risk, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to leverage. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ $(1 + \text{rate})$. Steps: Tk 100,000 $\times$ $(1 + 0.10)$ = Tk 110,000. Answer: Tk 110,000.

This section builds the idea of total leverage and risk before moving to application.

Earnings per share. Earnings left for owners of common stock divided by the number of shares..

The term earnings per share gives the chapter a precise name for this part of the problem.

Preferred stock. Shares that pay a fixed dividend, which must be paid before any dividend on common stock..

The term preferred stock gives the chapter a precise name for this part of the problem.

The central question in total leverage and risk is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

The second pass checks both the arithmetic and the financial interpretation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

SUMMARY

The central purpose of Risk and Return is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Daniel Bernoulli, 1738 paper on measuring risk, which showed that people judge a gamble by its meaning to them as well as by its average. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

09.T1 define risk and say how business risk differs from financial risk, using a new firm

09.T2 compute the expected value, then the variance, the standard deviation and the coefficient of variation, for a new two-asset table and state which is riskier per unit of return

09.T3 explain standard deviation in plain words

In this chapter, **Variable cost** means A cost that rises and falls with the number of units sold, such as the price paid for each book. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For costs and the break-even point, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

10.2 OPERATING LEVERAGE

The purpose here is to make operating leverage concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Contribution margin** means Selling price per unit minus variable cost per unit; it pays first for fixed costs, then gives profit. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Break-even point** means The sales volume at which total revenue equals total cost, so profit is zero. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **EBIT** means Earnings before interest and tax: sales minus operating costs. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Variable cost. A cost that rises and falls with the number of units sold, such as the price paid for each book..

The term variable cost gives the chapter a precise name for this part of the problem.

The central question in costs and the break-even point is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of operating leverage before moving to application.

Contribution margin. Selling price per unit minus variable cost per unit; it pays first for fixed costs, then gives profit..

The term contribution margin gives the chapter a precise name for this part of the problem.

Break-even point. The sales volume at which total revenue equals total cost, so profit is zero..

The term break-even point gives the chapter a precise name for this part of the problem.

EBIT. Earnings before interest and tax: sales minus operating costs..

The term ebit gives the chapter a precise name for this part of the problem.

The central question in operating leverage is what changes when the underlying decision changes.

chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Accounts payable** means Money a business owes its suppliers for goods already received. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Operating cycle** means The days from buying stock to collecting cash from its sale. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Cash conversion cycle** means The operating cycle minus the days a business takes to pay its suppliers. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Float** means Money tied up in payments that are sent but not yet usable, such as a cheque in the post. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For cycles, float and estimating needs, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

WORKED EXAMPLE

Consider Meghna and Sons, the running business used in this course. The firm must make a decision related to working capital and short-term financing. Start by listing the known amounts and their dates. Separate cash inflows from cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Accounts payable. Money a business owes its suppliers for goods already received..

The term accounts payable gives the chapter a precise name for this part of the problem.

Operating cycle. The days from buying stock to collecting cash from its sale..

The term operating cycle gives the chapter a precise name for this part of the problem.

Cash conversion cycle. The operating cycle minus the days a business takes to pay its suppliers..

The term cash conversion cycle gives the chapter a precise name for this part of the problem.

Float. Money tied up in payments that are sent but not yet usable, such as a cheque in the post..

The term float gives the chapter a precise name for this part of the problem.

The central question in cycles, float and estimating needs is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

A complete worked example joins the chapter ideas in one decision.

cash outflows, identify the rate or comparison rule required by the problem, and keep the units consistent. If a formula is needed, write it before substituting numbers. This prevents a familiar-looking number from being placed in the wrong part of the expression.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ (1 + rate). Steps: Tk 100,000 $\times$ (1 + 0.10) = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

In this chapter, **Net working capital** means Current assets minus current liabilities; what remains for trading after short debts are paid. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Matching principle** means Pay for short-lived needs with short-term money and for lasting needs with long-term money. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Short-term financing** means Money borrowed to be repaid within one year. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Intermediate-term financing** means Money borrowed to be repaid in more than one year but not more than about five. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For working capital and how to finance it, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

14.2 SOURCES OF SHORT-TERM FINANCE

The purpose here is to make sources of short-term finance concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

Net working capital. Current assets minus current liabilities; what remains for trading after short debts are paid..

The term net working capital gives the chapter a precise name for this part of the problem.

Matching principle. Pay for short-lived needs with short-term money and for lasting needs with long-term money..

The term matching principle gives the chapter a precise name for this part of the problem.

Short-term financing. Money borrowed to be repaid within one year..

The term short-term financing gives the chapter a precise name for this part of the problem.

Intermediate-term financing. Money borrowed to be repaid in more than one year but not more than about five..

The term intermediate-term financing gives the chapter a precise name for this part of the problem.

The central question in working capital and how to finance it is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of sources of short-term finance before moving to application.

before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

William Sharpe's 1964 paper on capital asset prices, and the 1990 Nobel Prize in economics shared by Markowitz, Miller and Sharpe. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

SOURCES

Chapter 9, 'Risk and Return', section 9.3 'Measuring risk' — standard deviation as the risk measure; Chapter 13, 'Valuation of Shares', section 13.3 'Valuing common stock' — required return used as the discount rate; Chapter 15, 'Long-Term Financing and the Capital Market', section 15.4 'The cost of capital' — equity cost..

REVIEW QUESTIONS

11.T1 explain risk premium and why investors demand it, with new figures

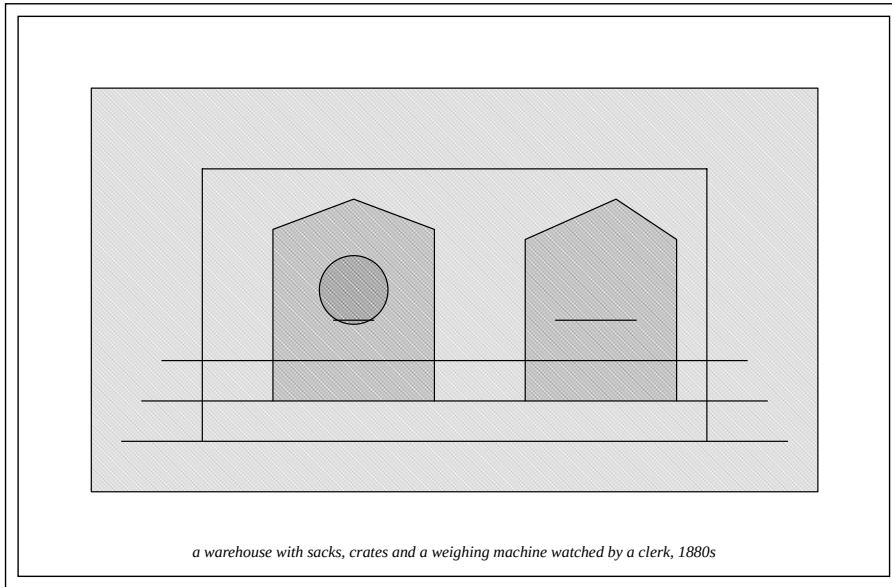
11.T2 separate systematic from unsystematic risk using a new list of risks

11.T3 compute required return for three new betas

11.T4 describe the three forms of market efficiency and what each implies

PAUSE AND CHECK 11.P1 After risk premium, state the rule in your own words.

PAUSE AND CHECK 11.P2 After beta, state the rule in your own words.



CHAPTER XIV

Working Capital and Short-Term Financing

A practical question about working capital and short-term financing



This chapter introduces working capital and short-term financing for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

This section builds the idea of the idea of valuation before moving to application.

Valuation. Finding what an asset is worth today by adding the present values of the cash it will bring..

The term valuation gives the chapter a precise name for this part of the problem.

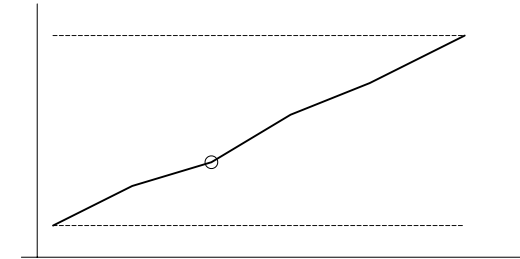


FIG. 12.1. Valuation of Bonds figure 1

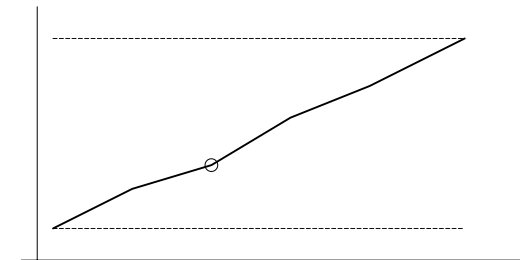


FIG. 12.2. Valuation of Bonds figure 2

12.1 THE IDEA OF VALUATION

The purpose here is to make the idea of valuation concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Valuation** means Finding what an asset is worth today by adding the present values of the cash it will bring. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

SUMMARY

The central purpose of Valuation of Shares is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

The South Sea Company of 1720: its share price rose and collapsed in one year while its earnings stayed small. Valuation theory: John Burr Williams, *The Theory of Investment Value* (1938) and Myron Gordon (1959). The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

SOURCES

Chapter 12, 'Valuation of Bonds', section 12.1 'The idea of valuation' — the principle used; Chapter 11, 'Required Return, CAPM and Market Efficiency', section 11.3 'The capital asset pricing model' — the required return; Chapter 15, 'Long-Term Financing and the Capital Market', section 15.2 'Debt and equity as sources' — stock as a source of money..

REVIEW QUESTIONS

13.T1 say how common stock differs from a bond using a new firm

A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

The chapter points back to related ideas elsewhere in the book.

The central question in what a bond is is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of pricing a bond before moving to application.

Premium bond. A bond that sells for more than its face value..

The term premium bond gives the chapter a precise name for this part of the problem.

Discount bond. A bond that sells for less than its face value..

The term discount bond gives the chapter a precise name for this part of the problem.

Semiannual interest. Interest paid twice a year, each payment being half the yearly amount..

The term semiannual interest gives the chapter a precise name for this part of the problem.

The central question in pricing a bond is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For what a bond is, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

12.3 PRICING A BOND

The purpose here is to make pricing a bond concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Premium bond** means A bond that sells for more than its face value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Discount bond** means A bond that sells for less than its face value. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Semiannual interest** means Interest paid twice a year, each payment being half the yearly amount. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For pricing a bond, keep the decision in view. Ask what the business is trying to buy,

* * *

13.2 VALUING PREFERRED STOCK

The purpose here is to make valuing preferred stock concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Capital gain** means The profit from selling an asset for more than its purchase price. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Perpetuity** means A series of equal payments that continues for ever. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For valuing preferred stock, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

13.3 VALUING COMMON STOCK

The purpose here is to make valuing common stock concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Dividend valuation model** means A way of valuing a share as the present value of the dividends it is expected to pay. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

This section builds the idea of valuing preferred stock before moving to application.

Capital gain. The profit from selling an asset for more than its purchase price..

The term capital gain gives the chapter a precise name for this part of the problem.

Perpetuity. A series of equal payments that continues for ever..

The term perpetuity gives the chapter a precise name for this part of the problem.

The central question in valuing preferred stock is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of valuing common stock before moving to application.

Dividend valuation model. A way of valuing a share as the present value of the dividends it is expected to pay..

The term dividend valuation model gives the chapter a precise name for this part of the problem.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

British consols, perpetual bonds first issued in the 1750s, and the UK government's redemption of its 3.5% War Loan in 1915. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

12.T1 explain what a bond is and name its main features with a new issue

12.T2 compute the value of three new bonds paying interest yearly and say whether each is at a premium, discount or par

12.T3 explain the link between coupon rate, required return and price

12.T4 define yield to maturity in plain words

PAUSE AND CHECK 12.P1 After bond features, state the rule in your own words.

PAUSE AND CHECK 12.P2 After pricing, state the rule in your own words.

the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

The second pass checks both the arithmetic and the financial interpretation.

A comparison is useful only when both alternatives are measured on the same basis.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

Given: a starting amount of Tk 100,000, a one-period rate of 10%, and a one-period horizon. Find: the ending value. Formula: Future value = Present value $\times$ $(1 + \text{rate})$. Steps: Tk 100,000 $\times$ $(1 + 0.10)$ = Tk 110,000. Answer: Tk 110,000. Check: the increase is Tk 10,000, which is 10% of Tk 100,000, so the result is internally consistent.

Suppose an alternative offers Tk 115,000 after one period. The difference from the first outcome is Tk 5,000. That does not by itself make the second choice preferable. We still need to know whether the second amount has the same risk, timing, required return, financing terms, and business consequences. Finance begins with a numerical comparison, then asks whether the comparison is fair.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

The chapter has connected its main terms, method, and business decision.

Finance becomes stronger when a number is followed by a question about meaning.

Most errors arise from mixing terms, timing, signs, or decision rules.

A historical case shows how the chapter's financial problem appeared in practice.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

SUMMARY

The central purpose of Working Capital and Short-Term Financing is to help a beginner identify the relevant financial quantities and use them consistently. The definitions in the chapter give each idea a fixed meaning. The worked example shows the expected order: identify what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

IN HISTORY

Toyota's just-in-time system, developed from the late 1940s by Taiichi Ohno and others to cut stock held. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

REVIEW QUESTIONS

- 14.T1 say what working capital is and why a profitable firm can still run short
- 14.T2 compute the cost of giving up a cash discount for new terms and recommend the cheaper source
- 14.T3 find the cash conversion cycle and the financing need for new data
- 14.T4 define float and its parts in plain words

APPENDIX A

Maths You Will Use



what is given, find what is required, choose the formula or rule, work through the steps, state the answer, and check the result.

THINK IT THROUGH

Ask what the number says about timing, risk, cash, value, or financing. Then ask what information could change the conclusion. A good financial explanation does not hide assumptions. It names them, uses the stated method, and distinguishes a numerical result from a business judgement.

Finance becomes stronger when a number is followed by a question about meaning.

WHERE PEOPLE GO WRONG

Check whether a cash flow is an inflow or an outflow, whether a percentage is written as a decimal before calculation, and whether amounts occur at the same date. Check units and periods. If a result looks surprising, inspect the inputs before changing the formula. The purpose of the check is to catch a mistake, not to force a preferred answer.

Most errors arise from mixing terms, timing, signs, or decision rules.

IN HISTORY

United States Leasing Corporation, founded in San Francisco in 1952, an early firm of the modern equipment-leasing industry. The historical point is not that the old setting was identical to a modern firm. It is that people faced a recognisable problem involving money, timing, ownership, risk, valuation, or control. Modern finance gives that problem a more explicit vocabulary and a set of methods for comparing alternatives.

A historical case shows how the chapter's financial problem appeared in practice.

REVIEW QUESTIONS

16.T1 say what a lease is and compare operating and financial leases with a new asset

16.T2 list the steps of the lease-or-buy comparison

16.T3 compute the after-tax present value of a new lease and purchase

16.T4 explain the tax shield of interest and depreciation

PAUSE AND CHECK 16.P1 After kinds of lease, state the rule in your own words.

PAUSE AND CHECK 16.P2 After after-tax outflows, state the rule in your own words.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

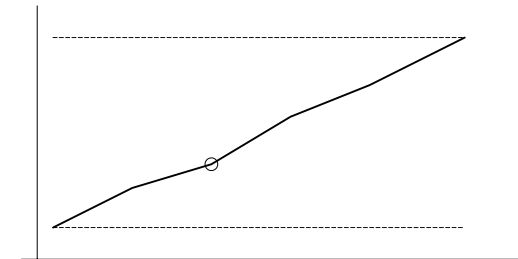


FIG. 15.1. Long-Term Financing and the Capital Market figure 1

15.1 TERM LOANS AND THE FEATURES OF FINANCING

The purpose here is to make term loans and the features of financing concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Term loan** means A loan repaid in regular instalments over a fixed number of years, usually from a bank. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Collateral** means An asset a lender may take if the borrower fails to repay. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Covenant** means A condition in a loan agreement that the borrower promises to keep. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

This section builds the idea of term loans and the features of financing before moving to application.

Term loan. A loan repaid in regular instalments over a fixed number of years, usually from a bank..

The term term loan gives the chapter a precise name for this part of the problem.

Collateral. An asset a lender may take if the borrower fails to repay..
The term collateral gives the chapter a precise name for this part of the problem.

Covenant. A condition in a loan agreement that the borrower promises to keep..

The term covenant gives the chapter a precise name for this part of the problem.

The central question in term loans and the features of financing is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

16.2 KINDS OF LEASE

The purpose here is to make kinds of lease concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Operating lease** means A short lease, cancellable, where the lessor keeps most of the risks of owning the asset. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Financial lease** means A long lease, not cancellable, that covers most of the asset's life and works like a loan. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Sale-and-leaseback** means A deal in which a firm sells an asset it owns and leases it back from the buyer. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For kinds of lease, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

16.3 THE LEASE-OR-BUY DECISION

The purpose here is to make the lease-or-buy decision concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the

This section builds the idea of kinds of lease before moving to application.

Operating lease. A short lease, cancellable, where the lessor keeps most of the risks of owning the asset..

The term operating lease gives the chapter a precise name for this part of the problem.

Financial lease. A long lease, not cancellable, that covers most of the asset's life and works like a loan..

The term financial lease gives the chapter a precise name for this part of the problem.

Sale-and-leaseback. A deal in which a firm sells an asset it owns and leases it back from the buyer..

The term sale-and-leaseback gives the chapter a precise name for this part of the problem.

The central question in kinds of lease is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of the lease-or-buy decision before moving to application.

15.3 RAISING FUNDS IN THE CAPITAL MARKET

This section builds the idea of raising funds in the capital market before moving to application.

Initial public offering. A company's first sale of shares to the public..

The term initial public offering gives the chapter a precise name for this part of the problem.

Rights issue. An offer of new shares to existing shareholders, in proportion to their holdings..

The term rights issue gives the chapter a precise name for this part of the problem.

Underwriter. An institution that promises to buy any shares of a new issue that the public does not..

The term underwriter gives the chapter a precise name for this part of the problem.

The central question in raising funds in the capital market is what changes when the underlying decision changes.

A small numerical example shows how the idea works before a more realistic case.

The business meaning matters as much as the arithmetic.

This section builds the idea of the cost of capital before moving to application.

The purpose here is to make raising funds in the capital market concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.

In this chapter, **Initial public offering** means A company's first sale of shares to the public. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Rights issue** means An offer of new shares to existing shareholders, in proportion to their holdings. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

In this chapter, **Underwriter** means An institution that promises to buy any shares of a new issue that the public does not. The important point is to use the term consistently. If the same business fact is described with a different everyday word, the calculation does not change, but the financial meaning can become less precise.

Begin with the cash or value that is actually being considered. Then identify its timing, its size, and the uncertainty around it where relevant. Next ask which comparison is valid. A financial rule is useful because it converts a business question into quantities that can be compared. The method is not a substitute for judgement; it gives judgement a disciplined base.

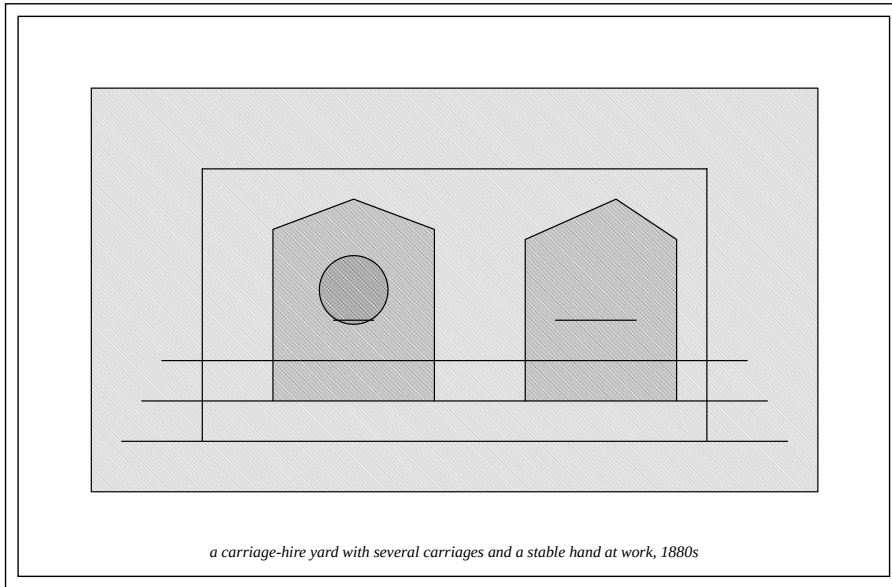
Worked example: suppose a firm has a starting amount of Tk 100,000 and the relevant rate is 10% for one period. A one-period increase is Tk 10,000, so the ending amount is Tk 110,000. If the timing changes, the comparison changes too. The same discipline applies when the numbers are larger: state what is given, state what must be found, choose the formula, show the steps, state the answer, and check whether the result is sensible.

A calculation can be correct and still answer the wrong question. For raising funds in the capital market, keep the decision in view. Ask what the business is trying to buy, finance, protect, compare, or value. Then check the direction of the result. If a larger cost is treated as a benefit, or a later cash flow is treated as though it arrives today, the arithmetic can hide the error.

* * *

15.4 THE COST OF CAPITAL

The purpose here is to make the cost of capital concrete. A beginner should be able to name the problem, identify the quantities that matter, and explain why the method fits the problem. The discussion follows the chapter sequence rather than assuming earlier knowledge beyond the stated need.



CHAPTER XVI

Leasing

A practical question about leasing



This chapter introduces leasing for a first-year reader. It builds each idea from a familiar business problem, defines the terms before using them, and connects the calculations to decisions.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A short restatement keeps the financial meaning attached to the calculation.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

A useful check is to restate the decision without the formula. Say what the business owns, what it owes, what cash arrives, what cash leaves, and when each amount occurs. Then ask whether the comparison uses the same period and the same units. This habit makes later chapters easier because each method begins with a disciplined description of the cash or value being studied.

W

Wealth maximisation, 18–20, 24

Weighted average cost of capital, 121, 124

Working capital, 108–109, 111, 114

Y

Yield to maturity, 93, 98

Cash inflow. Money coming into a business or to an investor, shown as a positive number. (*Chapter V*)

Cash outflow. Money going out of a business or from an investor, shown as a negative number. (*Chapter V*)

Coefficient of variation. Standard deviation divided by expected value, giving risk per unit of expected return. (*Chapter IX*)

Collateral. An asset a lender may take if the borrower fails to repay. (*Chapter XV*)

Common stock. Shares that give owners a vote and a claim on profit after all other claims are met. (*Chapter XIII*)

Company. A business set up as a legal person of its own, owned through shares, so the owners are separate from the business. (*Chapter I*)

Compound interest. Interest calculated on the principal sum and also on interest already earned. (*Chapter IV*)

Compounding period. The length of time after which interest is added to the balance, such as a year or six months. (*Chapter IV*)

Constant growth model. The dividend valuation model for dividends that grow at the same rate every year. (*Chapter XIII*)

Contribution margin. Selling price per unit minus variable cost per unit; it pays first for fixed costs, then gives profit. (*Chapter X*)

Controller. The officer in charge of accounting records, reports and checking that money is used as planned. (*Chapter II*)

Cost of capital. The return a firm must earn on its investments to satisfy those who supply its money. (*Chapter XV*)

Coupon rate. The yearly interest of a bond, as a percentage of its face value. (*Chapter XII*)

Covenant. A condition in a loan agreement that the borrower promises to keep. (*Chapter XV*)

Debenture. A bond backed only by the issuer's general promise to pay, not by a particular asset. (*Chapter XV*)

Depreciation. The share of an asset's cost charged against income each year as the asset wears out. (*Chapter VII*)

Discount bond. A bond that sells for less than its face value. (*Chapter XII*)

Discount rate. The yearly percentage used in discounting; it states the return you could earn elsewhere. (*Chapter V*)

Discounted payback period. The payback time when each cash flow is first discounted to present value. (*Chapter VIII*)

Discounting. Finding present value by removing the interest a sum could earn before it arrives. (*Chapter V*)

Diversification. Spreading money across many investments so that one failure does less damage. (*Chapter XI*)

Dividend. A share of a company's profit paid out to its owners instead of being kept in the business. (*Chapter II*)

Dividend decision. A choice about how much profit to pay out to owners and how much to keep for the business. (*Chapter II*)

Inventory, 110

Investment decision, 10

L

Lease, 126–128, 131

Lessee, 126

Lessor, 126–127

Leverage, 76–79, 81–82

Liability, 12

Long-term financing, 90, 105, 110, 118, 121, 123–124

M

Market return, 86

Matching principle, 109, 115

Maturity, 93, 98

Mixed stream, 35–36, 40

Mutually exclusive projects, 50

N

Net present value, 58–59, 61, 63–64

Net working capital, 109

O

Operating cash flow, 51, 56

Operating cycle, 111

Operating efficiency, 3

Operating lease, 127

Operating leverage, 77–78, 82

Opportunity cost, 26, 32

Ordinary annuity, 43, 48

P

Partnership, 3, 8

Payback period, 58–59

Perpetuity, 101, 106

Portfolio, 85

Preferred stock, 79, 101, 106

Premium bond, 94

Present value, 4, 13, 21, 24, 29, 34, 36, 39–40, 43–45, 48, ...

Principal, 19–20, 27–28

Principal sum, 27–28

Probability distribution, 67, 73

Profit, 2–3, 8, 10–11, 18–19, 24, 60, 64, 77, 100–101, 114, ...

Profit maximisation, 18–19, 24

Profitability index, 60, 64

Interest. The payment a borrower makes to a lender for the use of money, or the reward a saver earns. (*Chapter IV*)

Interest rate. The interest for one period written as a percentage of the sum, for example 8% a year. (*Chapter IV*)

Intermediate-term financing. Money borrowed to be repaid in more than one year but not more than about five. (*Chapter XIV*)

Internal rate of return. The discount rate at which a project's net present value is exactly zero. (*Chapter VIII*)

Inventory. The goods a business holds for sale, such as the books on a shop's shelves. (*Chapter XIV*)

Investment decision. A choice about which assets to buy, such as stock, a van or a new shop, in order to earn more money. (*Chapter II*)

Lease. A contract in which the owner of an asset lets another party use it for regular payments. (*Chapter XVI*)

Lessee. The user who pays to use the asset under a lease. (*Chapter XVI*)

Lessor. The owner who lets the asset out under a lease. (*Chapter XVI*)

Leverage. The use of a fixed cost or fixed payment so that a small change in sales causes a larger change in earnings. (*Chapter X*)

Liability. Money a business owes to others, such as a bank loan or an unpaid supplier bill. (*Chapter II*)

Long-term financing. Money raised for more than about five years, or with no fixed repayment date. (*Chapter XIV*)

Market return. The average return of all investments in a market during a period. (*Chapter XI*)

Matching principle. Pay for short-lived needs with short-term money and for lasting needs with long-term money. (*Chapter XIV*)

Maturity. The date on which a bond or loan must be repaid. (*Chapter XII*)

Mixed stream. A series of cash flows of different sizes in different periods. (*Chapter V*)

Mutually exclusive projects. Projects of which only one can be chosen, because choosing one rules out the other. (*Chapter VII*)

Net present value. The present value of all a project's future cash flows minus its initial investment. (*Chapter VIII*)

Net working capital. Current assets minus current liabilities; what remains for trading after short debts are paid. (*Chapter XIV*)

Operating cash flow. The cash a project brings in each year after paying its cash costs and tax. (*Chapter VII*)

Operating cycle. The days from buying stock to collecting cash from its sale. (*Chapter XIV*)

Operating efficiency. How much useful output a business gets from the money and effort it uses. A shop that sells more per taka of stock is more efficient. (*Chapter I*)

Operating lease. A short lease, cancellable, where the lessor keeps most of the risks of owning the asset. (*Chapter XVI*)

Operating leverage. The effect of fixed operating costs on how EBIT responds to sales. Measured by the degree of operating leverage (DOL), the

INDEX

A

Accounts payable, 111
Accounts receivable, 110
After-tax cash outflow, 128
Agency cost, 20, 23
Agency problem, 19–20, 24
Agent, 19
Annuity, 42–44, 48
Annuity due, 44
Asset, 10, 12, 50–52, 72, 86–87, 90, 92, 101, 105, 109, ...

B

Balance sheet, 12, 15
Beta, 85–87, 90
Bond, 92–95, 97–98, 105, 119
Break-even point, 76–77, 82
Business risk, 67, 72

C

Capital asset pricing model, 86–87, 105
Capital budgeting, 50, 52, 55
Capital expenditure, 50
Capital gain, 101
Capital market, 90, 105, 118–121, 123
Cash conversion cycle, 111, 114–115
Cash discount, 110, 114
Cash flow, 2–4, 7–8, 11–13, 15, 19–20, 23, 27–29, 32, 35–36, 39, ...
Cash inflow, 4, 13, 20, 29, 35–36, 45, 52, 61, 69, 79, ...
Cash outflow, 4, 13, 20, 29, 36, 45, 52, 61, 69, 79, ...
Coefficient of variation, 68, 72
Collateral, 118
Common stock, 79, 90, 100–102, 105–106
Company, 4, 7–8, 10, 18, 93, 105, 120, 124
Compound interest, 27–28, 32
Compounding period, 28
Constant growth model, 102
Contribution margin, 77
Controller, 11, 15
Cost of capital, 90, 120–121, 124
Coupon rate, 93, 98
Covenant, 118

Risk premium. The extra return demanded for taking risk instead of a risk-free investment. (*Chapter XI*)

Risk-free rate. The return on an investment with no risk of loss, such as a short-term government security. (*Chapter XI*)

Sale-and-leaseback. A deal in which a firm sells an asset it owns and leases it back from the buyer. (*Chapter XVI*)

Security market line. The graph of the capital asset pricing model, with beta on one axis and required return on the other. (*Chapter XI*)

Semiannual interest. Interest paid twice a year, each payment being half the yearly amount. (*Chapter XII*)

Share price. The price at which one share of a company can be bought or sold today. (*Chapter III*)

Shareholder. A person who owns part of a company by holding its shares. (*Chapter III*)

Short-term financing. Money borrowed to be repaid within one year. (*Chapter XIV*)

Simple interest. Interest calculated only on the principal sum, so it is the same every period. (*Chapter IV*)

Sole proprietorship. A business owned and run by one person, who keeps all profit and answers for all debts. (*Chapter I*)

Stakeholder. Anyone affected by a firm's actions, such as staff, customers, lenders and the community. (*Chapter III*)

Standard deviation. The square root of the variance; it shows how far results typically stray from the expected value. (*Chapter IX*)

Sunk cost. Money already spent that cannot be recovered whatever you decide now. (*Chapter VII*)

Systematic risk. Risk that affects all investments, such as a recession; it cannot be diversified away. (*Chapter XI*)

Tax shield. The tax saved because a cost, such as interest or depreciation, reduces taxable income. (*Chapter XVI*)

Term loan. A loan repaid in regular instalments over a fixed number of years, usually from a bank. (*Chapter XV*)

Terminal cash flow. The cash received or paid when a project ends, such as the sale of scrap. (*Chapter VII*)

Time value of money. The idea that a taka received today is worth more than a taka received later. (*Chapter IV*)

Timeline. A line marking each period, with the cash flow written at the point where it occurs. (*Chapter IV*)

Total leverage. The combined effect of operating and financial leverage on how owners' earnings respond to sales. Measured by the degree of total leverage (DTL), which equals DOL times DFL. (*Chapter X*)

Trade credit. Credit from a supplier who lets a buyer pay some days after delivery. (*Chapter XIV*)

Treasurer. The officer in charge of handling cash, raising money and keeping relations with banks. (*Chapter II*)

reasoning should use the facts given in the question and keep timing and units consistent.

15.t4 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

15.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

15.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER XVI · LEASING

16.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

16.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

16.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

16.t4 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

16.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

16.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER XI · REQUIRED RETURN, CAPM AND MARKET EFFICIENCY

11.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

11.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

11.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

11.t4 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

11.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

11.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER XII · VALUATION OF BONDS

12.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

12.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

12.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

12.t4 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

12.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

12.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER XIII · VALUATION OF SHARES

13.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The

03.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

03.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

03.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER IV · INTEREST AND FUTURE VALUE

04.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

04.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

04.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

04.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

04.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER V · PRESENT VALUE AND DISCOUNTING

05.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

05.t2 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

05.t3 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The reasoning should use the facts given in the question and keep timing and units consistent.

05.p1 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

05.p2 State the rule introduced at that point, then give one sentence explaining why it fits the problem.

CHAPTER VI · ANNUITIES

06.t1 A suitable answer identifies the relevant financial idea, applies the chapter definitions, and shows the calculation where the question asks for one. The